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Writing Analytics and AI for Special Education: Preliminary results on students with Autism Spectrum Disorder

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Abstract. This article discusses the utilization of writing analytics in Special Education, with a particular focus on students with Autism Spectrum Disorder (ASD). Research increasingly supports the use of data mining and Artificial Intelligence (AI) to analyze and support students' writing processes, showcasing the potential of these systems to enhance student engagement and the accuracy of automated feedback. However, concerns persist regarding potential biases and ethical implications. The literature highlights limitations in applying Writing Analytics and AI to atypical students since most research and tools are designed with typical students in mind, reflecting societal biases. Autistic students often encounter challenges in writing performance due to factors such as rigid style, limited vocabulary, and difficulties expressing thoughts. This paper presents a study involving the analysis of 2643 essays from secondary education students, including a subset with ASD, using text-to-network tools and NLP analysis to compare texts and examine computational linguistics metrics and text mining patterns. Preliminary findings suggest the necessity for tailored evaluation and interventions for ASD students. While AI offers opportunities for personalized interventions, further research is essential to effectively adapt current tools for atypical students.

Keywords: Writing Analytics, Autism Spectrum Disorder, Special Education, Text Mining, Natural Language Processing

1 Introduction

The interest in different forms of Artificial Intelligence (AI) within academic literature has grown exponentially, particularly since 2015 [24]. It is important to notice that AI is not a singular entity, but an umbrella term encompassing a set of modeling capabilities that are expanding on a daily basis. In this context, this paper

explores Writing Analytics (WA) in the educational context, with preliminary results of its potential and limitations in Special Education with students with Autism Spectrum Disorder (ASD).

Previous research states that WA applications can have high accuracy in evaluation tasks [25] and improve students' engagement in the writing process [26]. However, the predominant focus of research and the development of tools and applications, tends to revolve around the needs of typical students. Students with learning disabilities, when compared to their typically achieving peers, exhibit reduced planning time, generate less coherent ideas, and dedicate less effort to revising for meaning and content [9]. Concerning autistic students, it's widely acknowledged that they often demonstrate writing performance below their expected intellectual capabilities [10]. This discrepancy is attributed to factors such as a more rigid writing style, limited vocabulary, challenges in expressing thoughts through writing, difficulties with theory of mind, and weaker coherence. This paper presents a study with 2643 secondary education essays, aiming to bridge this gap.

2 Writing Analytics

The term writing analytics emerged to describe a field of study that focuses on analyzing and assessing texts to understand writing processes and products in educational contexts [13]. Writing Analytics' primary objective is supporting students to improve their writing skills. Patout [20] highlights the importance of these systems in providing context-aware feedback and scaffolding improvements in writing skills. However, Roscoe [23] emphasizes the need for these systems to offer appropriate and formative feedback, suggesting the use of pedagogically-guided algorithms. Lee [14] further supports the long-term effectiveness of these systems, demonstrating their positive impact on writing development over time. Another approach that is of interest in the field is that of verifying how precise can technological advances be in automated essay scoring (AES), when compared with human raters. While some earlier works have found only modest correlations between automated and human scores in MOOC context, suggesting some validity, other researchers advocate better results, describing AES approaches that may provide more specific feedback to writers and may be relevant to other natural language computations, such as the scoring of short answers in comprehension or knowledge assessments [17]. More recently, with the advent of generative AI, it has been also demonstrated how the integration of generative artificial intelligence with computational linguistics measures can enhance text scoring accuracy [18].

In this context, Natural Language Processing (NLP) plays an important role, providing techniques for textual classification based on linguistic characteristics [17]. Among the various applications of these technologies, the assessment of the readability level and complexity of texts has a certain prominence [27]. However, fewer studies focus on the use of textual metrics to evaluate textual quality, especially in languages other than English. Even fewer studies focus on the use of data-based instruction (DBI) for children with disabilities, although some research has provided preliminary support for the effectiveness of a combined

research-based intervention and DBI approach in addressing the intensive writing needs of students, particularly those with disabilities [11].

3 Text Analytics in Portuguese

As this study focuses on the analysis of texts written in Portuguese, we found it relevant to explore previous research in which Computational Linguistics and Natural Language Processing have been employed for the analysis of Portuguese texts. One of these studies was carried out by Evers and Finatto [6]. The authors used machine learning and natural language processing to distinguish levels of proficiency in Brazilian Portuguese. The work showed how a set of metrics, obtained using Coh-Metrix-Port, demonstrated non-random behavior for the established categories. A decision tree capable of classifying works into these categories was implemented, however, demonstrating a low classification capacity. Martins [15] carried out a study on the correlation between indicators of textual complexity and academic performance. The results indicated that, in relation to lexical complexity, there was a positive correlation between the frequency of nouns and progression, and a negative correlation between the frequency of verbs and progression. With regard to syntactic complexity, the correlation varied according to the type of subordinate clause in the two sets of data analyzed. Oliveira et al. [19] used regression models based on conventional machine learning methods and pre-trained deep learning language models to automatically estimate the cohesion of essays in Portuguese and English. They analyzed 4,570 essays in Portuguese and 7,101 in English, finding that a deep learning model performed best, with a moderate Pearson correlation with human-rated cohesion scores. Cavalcanti et al. [4] sought to identify the most influential textual characteristics in feedback texts written by teachers, seeking to predict their quality. A classifier was trained and evaluated, in Portuguese, at various levels of feedback. The most important variable identified was the number of paragraphs, directly linked to the amount of corrective information intended for students. More recently, Meira et al. [16] conducted a study focusing on university students' texts. The findings revealed a substantial correlation between most metrics and teachers' assigned grades, providing insights into textual quality indicators.

4 Writing with ASD and AI Bias

The research presented in this paper distinguishes itself from previous studies by directing its focus specifically towards exploring how writing analytics can be effectively employed to support writing instruction for students with Autism Spectrum Disorder (ASD). Previous research has demonstrated that students with ASD present specific difficulties in written expression, including differences in text length, legibility, size of handwriting, writing speed, spelling, and general text structure [8]. Although the research on ASD, education and technology is growing, the majority of research is interested in early education and young

children [7]. Such research is focused mainly on support for handwriting, grammar and spelling [5]. Some studies delved in more intricate aspects of writing, though. For instance, Hilvert [10] presented a linguistic analysis of the expository writing abilities of school-age children with ASD in comparison to neurotypical children. The study revealed that, in contrast to their neurotypical peers, children with ASD produced shorter expository texts containing more grammatical errors, requiring additional assistance during the writing assessment. In that sense, it is possible to say that the vast majority of Writing Analytics and AI tools availables are strongly biased. WA and AI tools are highly derivative from NLP metrics and text analysis that have been carried with neurotypical students as a target audience for the tools. This is a case of both data and societal bias in AI [12]. In that sense, analysis and tools built without ASD students in mind may not be suited to the individual needs of those students.

5 Method

For this study, 2643 essays produced by 1416 secondary school students were collected and analyzed using basic NLP and Network Analysis. The essays were obtained through the support of RevisãoOnline's team, a peer review system to support the process of producing and reviewing argumentative and dissertative texts, particularly tailored to the ENEM examination (National Secondary School Exam). For this preliminary study, 12 texts written by 10 students with ASD were identified and compared with the remaining 2631 texts written by their typical peers. The students with ASD were formerly diagnosed as such.

Basic NLP was done to the whole corpus of essays, comparing students with ASD numbers and the general population. Besides, the texts were converted to a network of terms using Sobek [21]. This process was carried out so network analysis could be done to the corpus, allowing the use of graph theory and the study of the relationships between terms and the inference of text characteristics.

Due to the small sample of students with ASD, this paper will focus on general measurements of network analysis, such as mean degree, distance and transitivity of those networks. All data was processed using R, version 4.2.2.

6 Basic NLP Results

A few NLP metrics were selected to understand patterns in students' writing behavior.

Number of sentences: The general average was 10.17 sentences in each text, while the median was 10 sentences. Standard deviation was found to be 5.02. Three essays from students with ASD had the number of sentences above average.

Word Count: The group mean was 256.9 words, with a median of 261 words, and standard deviation of 119.81. Once again students with ASD had values somewhat distant from those.

Average Sentence Length: The mean of average sentence length across the group was 27.05 words, with a median of 24.625 words. Standard deviation was 12.95. On this metric, 3 essays written by students with ASD had above average values.

Sentiments Mean: Sentiment analysis was used to analyze the essays. Due to the argumentative nature of the texts, results were expected to be close to zero. In fact, the general sentiment mean was 0.0002, with median 0 and standard deviation 0.005.

Basic NLP Results showed that the autistic students in this study showed variations in their writing patterns when compared to averages from their neurotypical peers. However, these patterns are not easily identifiable and seem to be highly individual. Number of sentences, word count and sentence length appear to be the metrics with a more noticeable pattern, since in all 3 of those, 9 out of 12 essays had values below average. This could suggest a tendency of writing short and fewer sentences, and generally using words with a more negative connotation. However, this data is not sufficient for drawing strong conclusions since no definitive pattern was found.

7 Network Analysis Results

The following analysis was conducted through the conversion of essays to a web of terms using text mining tool Sobek [21].

Average Degree: This network measure indicates how many connections to different nodes there are in a specific node. A higher average degree suggests a more densely connected network of terms, indicating stronger associations between terms in the text. A lower average degree indicates a sparser network with fewer connections between terms. The average degree was found to be 3.04, with median value of 3.125 and standard deviation of 0.836. Five essays of students with ASD had average degree above general mean, the highest being 3.78, while 7 essays had values below mean, the lowest being 0.79

Average Distance: This metric refers to the shortest path between two nodes. It provides insights into the overall connectedness and information flow within the text. A lower average distance indicates that terms in the text are closer to each other, which suggests an interconnected chain of ideas in a text. The general average distance across networks was 2.12, with median 2.08 and standard deviation 0.6. Three essays of students with ASD had average distance above the general mean, the highest being 3.61, and the other 9 essays had values below mean, the lowest being 1.

Transitivity: Network transitivity refers to the probability that two nodes that are connected to a common node are also connected to each other, i.e. that

triangles are formed in the network. The measure can indicate the extent to which terms in the text are clustered together, with a higher transitivity indicating a more tightly clustered network and therefore a text with clustered and dense ideas, understanding the overall structure of the text network. General average transitivity was found to be 0.23, with median of 0.18 and standard deviation of 0.16. Five essays written by students with ASD had values above average, with the highest network transitivity being 1, while 7 essays written by students with ASD had below-average values, the lowest being 0.

General Network Analysis Results were similar to those of basic NLP analysis. It was evident that for the measurements selected (degree, distance and transitivity), students with ASD had a range of atypical values, but with no evident pattern could be traced. Network Average Distance was the metric with higher congruence, with 9 out of 12 essays with below-average values. More data is necessary to better understand and possibly generalize these results.

8 Conclusions and Future Directions

Natural Language Processing (NLP) and Network Analysis (NA) have shown potential in detecting atypical patterns in the writing styles associated with Autism Spectrum Disorder (ASD). While these tools successfully identify deviations from typical linguistic and structural norms, they fall short of providing a comprehensive understanding of the unique writing patterns specific to individuals with ASD. The complexity of ASD expression and individual's characteristics and inconsistencies poses a challenge for NLP and NA to pinpoint a singular direction in interpreting the ASD writing, and at the moment is not possible to verify if such singular direction exists. Many AI systems rely on measures derived from NLP and NA. In that sense, the identified atypicalities raise concerns regarding the suitability of existing systems for ASD students. The reliance on generic linguistic and structural measures may contribute to the generation of frustrating experiences for ASD students and unhelpful tools for educators, as these systems may not be able to accommodate their distinctive communication styles and limitations. To address this challenge, it is imperative to tailor AI systems explicitly to the nuanced needs of ASD students, ensuring that these technologies enhance rather than hinder their learning experiences. This would require further research and AI systems that do not rely on generalization of metrics as a measurement of writing success.

The current limitations in adapting AI systems for ASD students highlight a significant bias in the existing AI landscape. The predominant focus on samples from neurotypical writers in the development of AI systems has created a bias that may render these technologies less suitable for the distinctive needs of individuals with ASD and other atypical students, that are still to be fully understood. The lack of representation from ASD writers in training datasets may contribute to the difficulty of these systems in accurately capturing the nuances of atypical writing patterns. Recognizing and addressing this bias becomes paramount in the pursuit of developing inclusive and effective AI tools for individuals with ASD. Finally, it is essential to acknowledge the limitations of the current findings, primarily stemming from the

utilization of a small sample size of students with ASD and their essays. Generalizing the outcomes to a broader population of ASD students demands further research with larger samples. More extensive research is crucial to understand how AI systems can be effectively adapted to the highly individualized needs and limitations of atypical students.

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